

Uncertainty Reserve Design of the New York Power Grid with High Offshore Wind Resources

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Abstract— With the fast incorporation of offshore wind resources (OSW) into the power grid, reliable grid operation has been increasingly challenged by OSW uncertainties. In New York, this challenge is compounded by the strong spatial correlation of OSW production due to the proximity of planned OSW farms. Failing to consider this correlation in reserve planning can lead to over-procurement. To solve this challenge, this study proposes an easy-to-implement uncertainty reserve technology that incorporates inter-site correlation using a risk-based aggregation method. The uncertainty reserve production cost is estimated based on the historical operating reserve prices in New York State, and the potential impact of the uncertainty reserve from OSW on a representative New York Power Grid is assessed. Results show that up to 1,636 MW uncertainty reserves may need to be procured hourly, with costs reaching \$7,366 per hour to manage OSW uncertainty by Year 2035 in New York State.

Keywords— *forecast uncertainty, offshore wind, operating reserves, risk-based aggregation, uncertainty reserve*

I. INTRODUCTION

The U.S. is targeting the integration of 30 gigawatts (GW) of offshore wind (OSW) energy resources by the year 2030 and 110 GW by the year 2050 [1]. Achieving this rapid scale-up presents significant challenges for the economic and reliable operation of the power grid. These challenges are heightened by the large capacity of individual OSW projects—for example, Empire Wind 1 is rated at 800 MW—and the geographic clustering of multiple OSW farms. In New York State, for instance, the planned 9 GW of OSW capacity by 2035 will be concentrated within approximately 100 miles of the New York City/Long Island Bight. This close proximity results in shared weather conditions across farms, creating highly correlated offshore generation uncertainties. Addressing these challenges will require a fundamental redesign of operating reserve strategies to ensure the reliability of bulk power system operations.

Regional Transmission Organization /Independent System Operators (RTOs/ISOs) have traditionally designed operating reserve products to attenuate the negative impact from power grid contingencies and ensure reliable grid operation. These reserves are typically sized based on the single largest generation contingency. For example, in New York, NYISO procures 1,310 MW of 10-minute reserves to account for the potential loss of a large nuclear plant in the New York Control Area. However, current reserve procurement practices do not

explicitly account for renewable generation uncertainties. Recognizing this gap, RTOs/ISOs—including NYISO—have begun developing new reserve products, such as uncertainty reserves and dynamic reserves[2],[3], to address the variability and limited predictability of renewable energy sources. Despite these efforts, the specific impact of offshore wind (OSW) resources on uncertainty reserve requirements remains insufficiently studied. In particular, uncertainty reserves may need to be procured in addition to traditional contingency reserves, which could significantly increase reserve requirements. Given the co-optimization of energy and reserves in market operations, this could drive up both energy and reserve prices. Therefore, it is essential to design uncertainty reserve mechanisms tailored to OSW resources—mechanisms that support grid reliability under high OSW penetration while accounting for their interaction with existing static (contingency-based) reserve structures.

Due to the geographic clustering of offshore wind (OSW) farms, their power output tends to be highly correlated. Traditional operating reserve designs assume generation failures occur as the complete loss of a single unit. However, with OSW resources, correlated forecast errors and simultaneous production swings across multiple farms can create substantial supply shortfalls—posing a risk of load loss during grid operations. To address this, a dedicated “uncertainty reserve” framework is needed—one that explicitly accounts for the correlated nature of OSW generation. Grid operators must incorporate this correlation into reserve planning to ensure sufficient generation resources are available to manage the variability and uncertainty of OSW power. To bridge the aforementioned gaps, this study will

- 1) Develop and quantify the uncertainty reserve required to manage OSW uncertainties with correlation in reliable power grid operation.
- 2) Validate the reserve procurement strategy with the buoy measurement data in the New York City/Long Island Bight and forecasted wind generation to provide support in uncertainty reserve design.
- 3) Assess the potential impact of the uncertainty reserve on generation dispatch with a representative New York power grid.

The rest of the paper is organized as follows: Section II outlines the methodology for the uncertainty reserve design to manage OSW uncertainties in grid operation and production cost simulation of the New York power grid. Section III shows

a case study with offshore wind buoy measurement and a representative New York power grid. Conclusions are described in Section IV.

II. METHODOLOGY

Uncertainty reserve helps maintain reliable grid operation by addressing forecast inaccuracies in the power grid, including forecasting uncertainties of load [4] and renewable energy sources such as onshore and offshore wind. This study focuses on developing uncertainty reserves to manage forecast uncertainties of the expected OSW farms in New York State.

A. Uncertainty Reserve Calculation

The NYISO proposes [2] an uncertainty reserve requirement methodology that combines both long-term and short-term forecast error data to ensure a more accurate and responsive reserve calculation. The methodology uses an 80% weighting for the prior year's static forecast error data and a 20% weighting for the previous two months of rolling forecast error data. This balance allows the reserve to account for both historical trends and more recent deviations in forecasting accuracy. The formula is applied separately for both load and wind to account for their distinct behaviors. The uncertainty reserve requirement is calculated using the following equation:

$$UR = 80\% \times (P_{yr}^{NL} \times NLF) + 20\% \times (P_{2m}^{NL} \times NLF) + 80\% \times (P_{yr}^{WP} \times WPF) + 20\% \times (P_{2m}^{WP} \times WPF) \quad (1)$$

where UR is the uncertainty reserve requirement; P_{yr}^{NL} is the prior year net load forecast error percentage; P_{2m}^{NL} is the rolling 2-month net load forecast error percentage; NLF is the net load forecast; P_{yr}^{WP} is the prior year wind forecast bin requirement percentage; P_{2m}^{WP} is the rolling 2-month wind forecast bin requirement percentage; and WPF is the wind power forecast.

The uncertainty reserve can be separately determined for each uncertain source, e.g., the load demand, land-based wind, offshore wind, and solar energy resources. This study focuses on designing the uncertainty reserve for New York state planned 9 GW OSW capacity by 2035 aligned with New York's Climate Leadership and Community Protection Act (CLCPA). The leased project areas (orange circles) are shown in Fig. 1(a). A well-known challenge is the limited availability of wind measurements in the North Atlantic OSW development regions (red stars). To demonstrate the reserve requirement, wind speed measurements from Buoy E05 in 2020 have been analyzed as an example.

To convert wind speed into OSW wind generation, we adopt the industry practice outlined by the NYISO [5], shown in Fig. 1 (b) where wind speed is translated into a percentage of nameplate power using turbine specifications. In our calculation, we follow the industry practice and assume the wind speed patterns observed in 2020 will remain consistent in future years, such as 2030 and 2035, with the installed OSW plant capacity changing according to New York State's policy goals. This approach allows us to calculate the wind power forecast error distribution as illustrated in Fig. 1 (c) for any

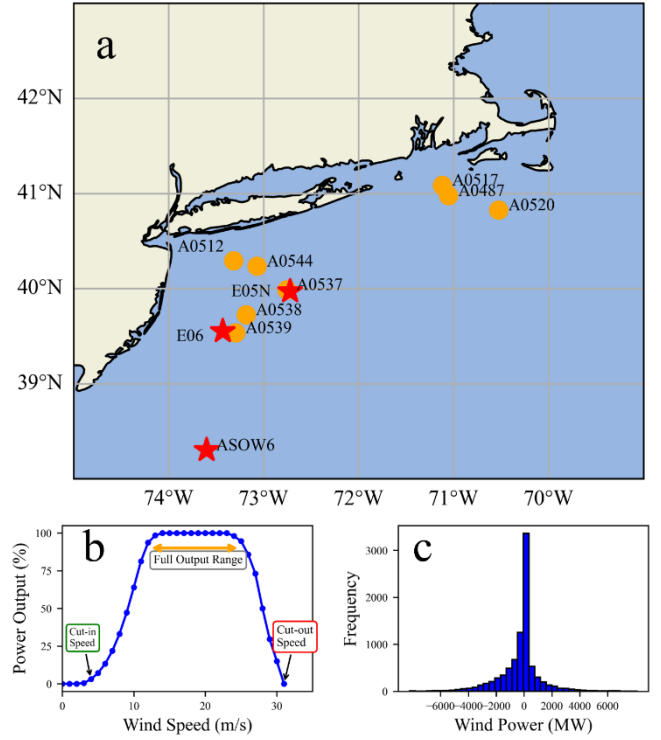


Fig. 1. a) Offshore wind leased area; b) Wind speed to wind power conversion curve; c) Wind power forecast error distribution.

given timestamp. By analyzing this distribution, we can determine the bin requirements percentage P_{yr}^{WP} and P_{2m}^{WP} for a desired percentile, which is then applied in the uncertainty reserve calculation in (1) for OSW resources.

B. Correlation Analysis of Offshore Wind Sites

This proposed study uses measurement data from two buoys deployed by the New York State Energy Research and Development Authority (NYSEDA) at Hudson South E06 and Hudson North E05N. From these buoys, actual wind speed data with a resolution of every 10 minutes [6] is collected at 158-meter altitude. For wind forecasts at any location, we use the RU-WRF model, a numerical weather forecasting engine, which has been developed to support offshore wind development in the mid/north Atlantic regions and is independently validated by the NREL [7].

Table 1 presents the offshore wind projects planned in New York State, including their respective capacities and developers. To model future deployment, we align with the targets outlined in New York State's Climate Leadership and Community Protection Act (CLCPA), which mandates 9,000 MW of offshore wind capacity by 2035. Year-by-year capacity projections are informed by the NYISO System & Resource Outlook study, specifically the Policy Case S1 scenario[8]. Capacities are matched as closely as possible to this reference trajectory for modeling consistency.

The forecasted wind power generation data from each leased offshore wind site was used to calculate correlation coefficients, with overall correlations shown as a function of distance in Fig. 2. High correlations were observed between geographically proximate sites. As an example, sites A0512

Table 1. Offshore Wind Lease Areas in New York State

Offshore Wind Development Projects			
Lease Area	Project Name	2030 Capacity (MW)	2035 Capacity (MW)
A0487	Sunrise Wind	924	924
A0512	Empire Wind	2076	2076
A0517	South Fork Wind	130	130
A0520	Beacon Wind	1230	1230
A0537	Bluepoint Wind	-	1700
A0538	Attentive Energy	1404	1404
A0539	Community Offshore Wind		1314
A0544	Vineyard Mid-Atlantic		1314
Total Capacity		5764	10092

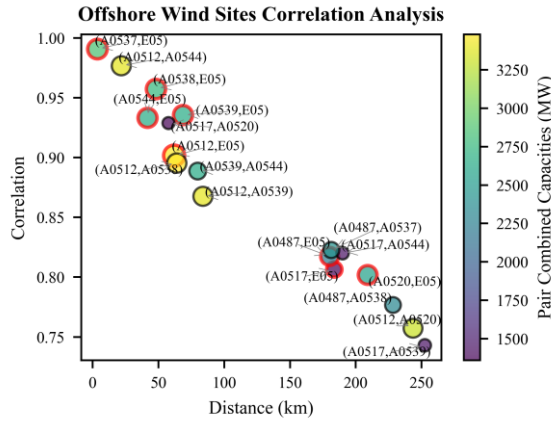


Fig. 2. Correlation coefficients for planned offshore wind projects as a function of the distance between sites.

and A0544 exhibit a strong correlation of 0.98, indicating highly similar generation patterns. The E05 pair (highlighted with a red circle in the plot) serves as a real observation data point. It is observed that correlation between sites decreases as distance increases, largely due to varying local meteorological conditions. When aggregating reserve requirements across multiple OSW sites, current practices often treat them as independent sources and sum individual reserves. However, this approach overlooks the correlation structure and thus the diversification benefit, potentially leading to an overestimation of reserve needs and incurring unnecessary system costs.

To quantify the aggregated uncertainty reserve requirement under correlated conditions, we adopt a correlation-based risk aggregation approach, commonly used in financial risk management for calculating the Solvency Capital Requirement (SCR) [9]. The total uncertainty reserve requirement is computed as follows:

$$UR_{total,Corr} = \sqrt{\sum_{i=1}^N \sum_{j=1}^N R_i \times R_j \times \rho_{ij}}, \forall i, j \in N \quad (2)$$

where, R_i & R_j are the uncertainty reserve requirements associated with plant i and plant j ; $\rho_{i,j}$ is the correlation between plant i and plant j ; and N is the number of plants. It

can be expressed as a quadratic form based on the reserve requirement vector $\mathbf{R}$ and correlation matrix $\boldsymbol{\rho}$ as

$$UR_{total,Corr} = \sqrt{\mathbf{R}^T \boldsymbol{\rho} \mathbf{R}} \quad (3)$$

This formulation captures not only the plant-specific uncertainty reserve, but also their interdependencies among plants represented by correlation matrix $\boldsymbol{\rho}$. Since $\boldsymbol{\rho}$ is positive semi-definite (PSD) by definition, the quadratic form $\mathbf{R}^T \boldsymbol{\rho} \mathbf{R} \geq 0$. This ensures $UR_{total,Corr}$ is real and non-negative. When the risks are fully correlated ($\rho_{ij} = 1$), the formulation (2) ensures the highest aggregated reserve, reflecting a worst-case scenario. This is particularly important in wind generation systems, where geographically distributed turbines may follow similar weather patterns. In such cases, simultaneous low outputs can occur, and higher reserves are essential to ensure system reliability.

C. Production Cost Modeling in Day Ahead Market

Sets

$\mathcal{T}$: Time periods (hourly) within optimization horizon

$\mathcal{N}$: Generators set

$\mathcal{ST}$: Storage set

$\mathcal{RR}$: Regulation and operating reserve of all types

$PR_{i,rr}$: Generator capacity (%) for reserves or regulation

Indices

t : time $t \in \mathcal{T}$

i, s : generator $i \in \mathcal{N}$, storage $s \in \mathcal{ST}$

Parameters

$\mathcal{L}_t$: Load demand at time t

MC_i : Marginal cost to produce energy (\$/MWh)

G_i^{max} : Maximum generation capacity (MW)

R_{rr} : Regulation/ reserve requirement (MW)

U_t : Uncertainty reserve requirement (MW)

Variables

$g_{i,t}$: Energy generated (MWh) by generators

$q_{i,rr,t}$: Regulation/reserve from generator i (MW)

$q_{s,rr,t}$: Regulation/reserve from storage s (MW)

$pr_{rr,t}$: Regulation/reserve procurement penalty

$u_{i,t}$: Uncertainty reserve from generator i (MW)

$u_{s,t}$: Uncertainty reserve from storage s (MW)

The energy production cost model aims to find the lowest-cost solution to meet demand while ensuring enough reserve resources are procured for regulation, uncertainty, and operating reserve needs. To illustrate this approach, the production cost model with linear Unit Commitment is developed for a representative New York Power Grid [10] as follows:

$$\min \sum_i^N \sum_t^{\mathcal{T}} MC_i \cdot g_{i,t} \quad (4)$$

subject to

$$\sum_i g_{i,t} + \sum_s (p_t^d - p_t^e) = L_t, \forall i \in \mathcal{N}, \forall s \in \mathcal{ST}, t \in \mathcal{T} \quad (5)$$

$$q_{i,\{Reg\},t} \leq G_i^{max} \cdot PR_{i,\{Reg\}}, \forall i \in \mathcal{N}, t \in \mathcal{T} \quad (6)$$

$$q_{i,\{10minSpin\},t} \leq G_i^{max} \cdot PR_{i,\{10minSpin\}}, \forall i \in \mathcal{N}, t \in \mathcal{T} \quad (7)$$

$$q_{i,\{10minSpin\},t} + q_{i,\{30min\},t} \leq G_i^{max} * PR_{i,\{30min\}}, \forall i \in N, t \in \mathcal{T} \quad (8)$$

$$g_{i,t} + \sum_{rr} q_{i,rr,t} + u_{i,t} \leq G_i^{max}, \forall i \in N, t \in \mathcal{T} \quad (9)$$

In this model, (5) ensures that the total generation combined with net energy storage must satisfy the load demand at each time t . Each generation unit can provide a percentage ($PR_{i,rr}$) of its total capacity G_i^{max} for ancillary services, such as spinning and regulation reserves. We considered three types of reserves $\{10minSpin, 30minSpin, Reg\}$ in $\mathcal{RR}$. (6), (7), and (8) define the maximum allowable capacities for regulatory reserve, 10-minute spinning reserve, and 30-minute reserve, respectively. The Independent System Operator (ISO) may define the required reserve R_{rr} values directly. Finally, (9) governs the total capacity allocation for all generators including uncertainty reserve.

$$\sum_i q_{i,rr,t} + \sum_s q_{s,rr,t} + pr_{rr,t} \geq R_{rr}, \forall t \in \mathcal{T} \quad (10)$$

$$\begin{aligned} \sum_i q_{i,\{10minSpin\},t} + \sum_s q_{s,\{10minSpin\},t} + pr_{\{10minSpin\},t} \\ + \sum_i u_{i,t} + \sum_s u_{s,t} \\ \geq U_t + R_{\{10minSpin\},t} \quad \forall t \in \mathcal{T} \end{aligned} \quad (11)$$

$$\begin{aligned} \sum_i q_{i,\{10minSpin\},t} + \sum_i q_{i,\{30minSpin\},t} \\ + \sum_s q_{s,\{10minSpin\},t} \\ + \sum_s q_{s,\{30minSpin\},t} + \sum_i u_{i,t} \\ + \sum_s u_{s,t} + pr_{\{30min\},t} \\ \geq U_t + R_{\{30minSpin\},t} \quad \forall t \in \mathcal{T} \end{aligned} \quad (12)$$

At each time instance, the total reserve including uncertainty and spinning reserve and regulation must be scheduled. (10) ensure sufficient regulation reserve is procured. Uncertainty reserve is formulated into the 10 minute reserve requirement in (11) and (12). Energy storage modeling has been integrated into the optimization model. Please note this model intends to evaluate the impact of uncertainty reserve on generation dispatch and the production cost of uncertainty reserve is separately assessed with historical price data.

III. CASE STUDIES

In this study, we quantify the uncertainty reserve requirements for OSW generation, focusing on New York State's CLCPA targets with expected installation capacity of 5.76 GW by 2030 and 9 GW by 2035. To estimate future wind power production, we utilize 2020 wind speed data for both observed and forecasted conditions, assuming that wind profile characteristics remain consistent across future years. Only the installed capacity is adjusted to reflect year-specific projections. The required uncertainty reserve is calculated using expected site-specific OSW capacity data; this reserve production cost and its impact on generation dispatch are assessed on a representative New York State power grid model.

A. Uncertainty Reserve Requirement Analysis for OSW

Using the projected installed capacity and the proposed uncertainty reserve formulation, we compute hourly reserve requirements for the years 2030 and 2035. Seasonal comparisons, as illustrated in Fig. 3, reveal an increased reserve requirement corresponding to the expansion of offshore wind (OSW) capacity. In 2035, the spring season exhibits the highest uncertainty reserve requirement, with 2391 MW at the **P50** level, corresponding to 23.69% of the installed offshore wind capacity. This elevated requirement is primarily driven by increased wind variability during the spring months. In contrast, the summer season demonstrates the lowest reserve need, with 1134 MW at **P50**, representing 11.24% of the installed capacity. This seasonal pattern is consistent across all offshore wind lease areas, reflecting spatially varying but seasonally aligned reserve needs driven by localized wind variability. A similar seasonal pattern is observed in the 2030 scenario.

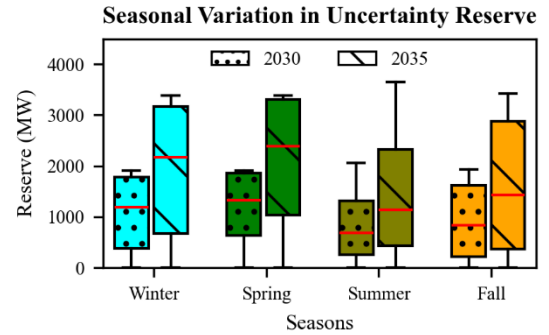


Fig. 3. Uncertainty reserve requirements in different seasons without considering spatial correlation.

Table 2. Seasonal Uncertainty Reserve Requirements at Different Quantiles for 2035

Season	Uncertainty Reserve MW at Different Quantiles					
	P25	P50	P75	P80	P90	P95
Winter	669	2178	3167	3267	3317	3363
Spring	1031	2391	3303	3361	3382	3382
Summer	435	1134	2326	2630	3216	3485
Fall	374	1425	2874	3095	3377	3382

While current reserve procurement strategies are typically based on static values throughout the year, our analysis underscores the importance of considering seasonal variability. Grid operators can leverage this information to flexibly select from multiple quantile levels (e.g., **P50**, **P75**, **P90**) to determine reserve margins that are both operationally robust and economically viable. Table 2 provides the expected uncertainty reserve MW by season for 2035 across the state, showing that reserve requirements increase with higher quantiles. Procuring reserves at higher quantiles can enhance system reliability and reduce the risk of loss-of-load events, but it may also introduce economic challenges due to increased operational costs. Some extreme days may require higher reserves, but most reserves are chosen near the median. Ultimately, reserve levels depend on grid operator's selection of percentiles of OSW uncertainties to balance between grid reliability and economics.

B. Spatial Correlation Consideration in Reserve Design

We calculate the uncertainty reserve requirement individually for each offshore wind plant and then aggregate these values to obtain a statewide reserve estimate. Also, we evaluate the spatial correlation among sites and apply the correlation-based aggregation approach using (2) to derive the total reserve considering inter-site dependencies. The results are summarized in Table 3. Incorporating spatial correlation leads to an approximate 8–9% reduction in average hourly uncertainty reserve requirements for both 2030 and 2035. This reduction reflects the benefit introduced by diversified offshore uncertainty.

Table 3. Average Hourly Seasonal Aggregated Uncertainty Reserve Requirements with and without Correlation Consideration (in MW)

Year	Winter	Spring	Summer	Fall
Aggregated Reserve with 100% Correlation				
2030	1077.91	1200.73	813.11	915.47
2035	1929.16	2120.92	1421.84	1606.35
Aggregated Reserve with Partial Correlation				
2030	987.09	1102.17	746.09	836.47
2035	1784.92	1965.32	1314.74	1480.98

C. Production Cost Analysis for OSW Uncertainty Reserve

We analyzed five years (2019–2023) of historical 10-minute spinning reserve price data across seasons as shown in Fig. 4, focusing on the New York City and Long Island load zones due to their geographic proximity to all planned offshore wind farms. From the resulting probabilistic hourly price distributions, the P50 values were extracted and used to estimate the cost of uncertainty reserve products.

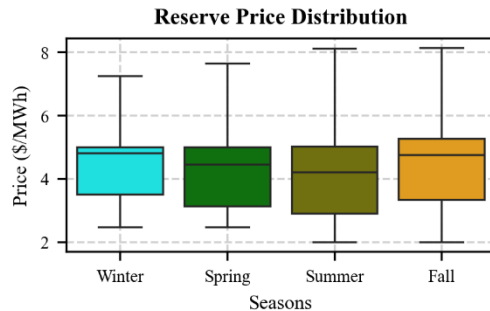


Fig. 4. Seasonal Distribution of 10-Minute Spinning Reserve P50 Prices Averaged Across New York City and Long Island Zones (2019–2023).

Table 4. Average Hourly Uncertainty Reserve and Annual Cost

Year	Average Reserve (MW)		Annual Cost (USD)	
	With 100% Correlation	With Partial Correlation	With 100% Correlation	With Partial Correlation
2030	1002	918	\$39,683,551	\$36,186,719
2035	1770	1636	\$70,133,596	\$64,527,926

To evaluate the economic implications of uncertainty reserve procurement, we estimate the annual production cost for both 2030 and 2035 using the P50 ancillary service price derived from historical NYISO market data (2019–2023).

Table 4 summarizes the results for aggregated reserve calculations with and without consideration of spatial correlation among offshore wind sites. Considering spatial correlation results in a notable reserve production cost reduction of \$3.50 million in 2030 and \$5.61 million in 2035 approximately. The reserve cost values are computed based on hourly reserve requirements and historical pricing. While this analysis does not predict future market behaviors, it provides critical insights into the potential cost impacts of uncertainty reserve procurement strategies, thereby informing the design and valuation of OSW uncertainty reserve products.

D. Impact on Generation Resource Dispatch from OSW Uncertainty Reserve

To evaluate the potential impact of the OSW uncertainty reserve on the generation resource dispatch, the representative New York Power Grid model as formulated in Section II was executed. The uncertainty reserve was added on top of the 10-minute static reserve requirement (1310MW) for each hour. Fig. 5 represents the resources to meet these reserve requirements throughout the day on December 31, 2035, a representative winter day. Only dispatchable generation resources were considered for providing uncertainty and operating reserves. Among these, battery storage plays a significant role in responding to uncertainty due to its fast-ramping capability and high flexibility.

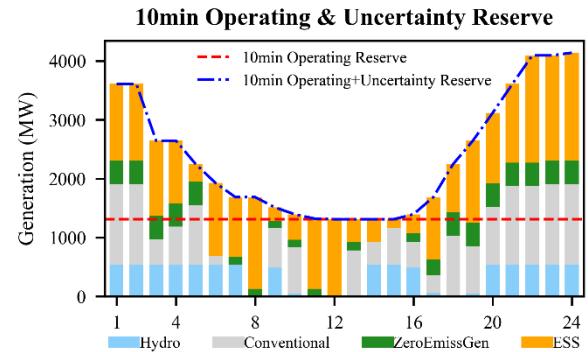


Fig. 5. Generation resources for 10 minutes regulating reserve and uncertainty reserve requirements.

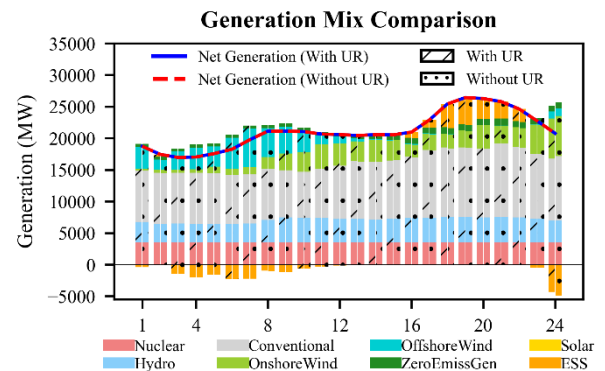


Fig. 6. Generation mixes with and without uncertainty reserve.

RTOs/ISOs are currently implementing an energy and ancillary service co-optimization framework. As such, the uncertainty reserve procurement will impact the generation

resources dispatch as demonstrated in Fig. 6. For the selected day, it is observed that net dispatched generation with and without the uncertainty reserve remains the same to meet the load demand while the dispatched generation resources are changed slightly. Notably, adjustments occur during peak and late-night hours, with fast-ramping resources such as natural gas units and storage systems adjusting their output more dynamically to maintain system balance.

Table 5. Generation comparison with and without uncertainty reserve consideration in production cost model.

Fuel	GasCC	GasCT	GasST	ZeroEmiss Gen	2-hr BESS
Without UR	8265	1058	822	972	(3705)*
With UR	8074	966	688	856	(3172)*

* Negative value represents the charging condition

In 2035, offshore wind capacity is projected to reach at least 9 GW and is assumed to be fully committed in production cost modeling to maximize renewable utilization. However, the inherent variability of OSW necessitates reserve support. For example, on December 31, OSW output reaches a peak of approximately 6 GW at hour 7 but drops sharply to just 1.1 GW by hour 24. The impact of this variability on generation mixes at hour 24 is detailed in Table 5, comparing dispatch results with and without uncertainty reserve. The table only lists resources with substantial variation in both scenarios. We can see that, without uncertainty reserves, natural gas units are utilized heavily to balance the load, while storage often holds surplus energy for future needs. With the uncertainty reserves, storage assets are mobilized more frequently, resulting in reduced dispatch demand on natural gas resources.

Most of the uncertainty reserve is met by energy storage systems as illustrated in Fig. 5, underlining their operational advantage in responding quickly to uncertainty events. Overall, the results demonstrate that uncertainty reserve procurement not only supports renewable integration but also improves the economic dispatch of generation resources by leveraging the flexibility of modern assets.

IV. CONCLUSION

This study examined the design of uncertainty reserve products as offshore wind penetration increases in New York State. Our analysis quantified reserve needs and their seasonal variations, emphasized the importance of incorporating spatial correlation in reserve design, estimated the costs of procuring uncertainty reserves, and assessed changes in the generation dispatch due to the integration of these reserves in grid operation. Results highlight significant contributions from high-ramping resources, such as natural gas and battery storage, to maintain reserve adequacy. Findings suggest that 1) a large uncertainty reserve quantity needs to be procured to manage the uncertainties of the expected OSW farms in New York State; 2) the geo-diversity of offshore wind farms contributes to the reduction of uncertainty reserve procurement to manage the offshore wind uncertainty; 3) it incurs high cost to procure uncertainty reserve required to manage OSW uncertainties, and 4) flexible resources will be critical to maintain grid reliability by balancing renewable variability and uncertainties.

ACKNOWLEDGMENT

This material is based upon work supported by the National Science Foundation under Grant No. 2338383 and Grant No. 2150238. The authors also greatly appreciate the discussion and support from the NYISO engineers. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation or NYISO.

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